

$K = 2$

Dataset	N	D	Deviation from ofv ABA [%]						cpu ABA [s]	Deviation from cpu ABA [%]			
			ofv ABA	P-N5	P-R5	P-R50	P-R500	Rand		P-N5	P-R5	P-R50	P-R500
Travel	5,454	24	130,871.97	0.0000	0.0000	0.0000	0.0000	-0.0032	0.002	39,878.1	30,187.8	38,414.2	97,237.8
Npi	10,440	40	87,153.62	0.0000	0.0000	0.0000	0.0000	-0.0123	0.004	241,508.5	31,899.7	42,924.1	120,903.4
Creditcard	30,000	24	719,976.00	0.0000	0.0000	0.0000	0.0000	-0.0043	0.012	68,143.9	31,143.8	40,420.8	90,756.6
Adult	32,561	110	3,581,598.08	-0.0010	0.0000	0.0000	0.0000	-0.0030	0.026	47,867.7	17,039.3	27,054.9	100,510.6
Plants	34,781	70	258,111.52	0.0000	0.0000	0.0000	0.0000	-0.0024	0.021	85,318.7	23,029.4	31,160.9	87,628.4
Bank	45,211	53	2,396,129.96	-0.0004	0.0000	0.0000	0.0000	-0.0012	0.026	59,364.8	25,085.0	33,778.6	90,131.1
Cifar10	50,000	3,072	9,520,051.68	0.0000	0.0000	0.0000	0.0000	-0.0016	0.501	821,214.7	9,806.5	68,128.7	648,435.3
Mnist	60,000	784	3,163,501.89	0.0000	0.0000	0.0000	0.0000	-0.0016	0.173	1,814,279.3	9,834.4	48,903.7	435,780.1
Survival	110,204	4	440,812.00	-0.0008	0.0000	0.0000	0.0000	-0.0006	0.034	42,578.3	40,809.8	52,535.4	85,187.8
Diabetes	253,680	22	5,580,938.00	0.0000	0.0000	0.0000	0.0000	-0.0004	0.702	17,777.2	4,923.0	6,304.4	13,489.9
Music	515,345	91	46,896,304.00	—	0.0000	0.0000	0.0000	-0.0001	0.493	—	15,467.0	24,821.9	113,499.6
Covtype	581,012	55	31,955,604.64	-0.0001	0.0000	0.0000	0.0000	0.0000	0.404	26,992.2	21,733.6	31,452.6	101,752.5
Imagenet8	1,281,167	192	13,327,181.18	—	0.0000	0.0000	0.0000	-0.0001	1.816	—	11,692.2	24,479.0	138,568.8
Imagenet32	1,281,167	3,072	265,674,710.38	—	—	—	—	-0.0001	13.760	—	—	—	—
Census	2,458,285	68	167,163,312.00	—	0.0000	0.0000	0.0000	0.0000	2.402	—	15,582.2	23,957.7	88,566.1
Finance	6,362,620	12	76,351,428.00	0.0000	0.0000	0.0000	—	0.0000	4.465	24,584.8	20,611.8	25,933.8	—
Average				-0.0002	0.0000	0.0000	0.0000	-0.0019		274,125.7	20,589.7	34,684.7	158,032.0

ABA is the contribution in this paper

$K = 2$

Dataset	N	D	Deviation from sd ABA [%]						range ABA [s]	Deviation from range ABA [%]				
			sd ABA	P-N5	P-R5	P-R50	P-R500	Rand		P-N5	P-R5	P-R50	P-R500	Rand
Travel	5,454	24	2.8	17,530.6	5,716.2	7,292.4	9,998.8	7,726.3	5.7	17,530.6	5,716.2	7,292.4	9,998.8	7,726.3
Npi	10,440	40	0.0	5,642.4	11,806.2	5,332.8	6,184.7	1,060,733.9	0.0	5,642.4	11,806.2	5,332.8	6,184.7	1,060,733.9
Creditcard	30,000	24	1,931.4	61.8	249.0	222.9	249.1	482.4	3,862.9	61.8	249.0	222.9	249.1	482.4
Adult	32,561	110	14,376.6	-27.6	30.9	6.1	26.2	55.7	28,753.3	-27.6	30.9	6.1	26.2	55.7
Plants	34,781	70	0.8	1,620.3	943.4	1,374.6	1,620.3	114,286.0	1.5	1,620.3	943.4	1,374.6	1,620.3	114,286.0
Bank	45,211	53	6,641.0	-39.1	-2.6	-0.3	-7.8	25.0	13,282.0	-39.1	-2.6	-0.3	-7.8	25.0
Cifar10	50,000	3,072	3.9	120,338.7	342,514.3	399,751.2	215,852.2	368,538.8	7.8	120,338.7	342,514.3	399,751.2	215,852.2	368,538.8
Mnist	60,000	784	5.9	5,119.2	338.0	7,516.0	3,231.4	15,569.9	11.8	5,119.2	338.0	7,516.0	3,231.4	15,569.9
Survival	110,204	4	1.1	51,118.0	26,483.7	19,260.4	10,691.0	51,999.6	2.1	51,118.0	26,483.7	19,260.4	10,691.0	51,999.6
Diabetes	253,680	22	9.6	10,461.8	4,229.8	6,520.1	15,076.8	32,382.0	19.2	10,461.8	4,229.8	6,520.1	15,076.8	32,382.0
Music	515,345	91	7,082.1	—	137.8	481.5	792.3	1,123.0	14,164.3	—	137.8	481.5	792.3	1,123.0
Covtype	581,012	55	93,118.9	-9.7	3.2	1.6	7.4	13.3	186,237.8	-9.7	3.2	1.6	7.4	13.3
Imagenet8	1,281,167	192	0.3	—	868,174.2	1,376,640.8	694,581.8	1,809,256.7	0.6	—	868,174.2	1,376,640.8	694,581.8	1,809,256.7
Imagenet32	1,281,167	3,072	7.9	—	—	—	—	736,724.5	15.9	—	—	—	—	736,724.5
Census	2,458,285	68	91.7	—	2,559.6	2,857.8	2,263.2	34,584.4	183.4	—	2,559.6	2,857.8	2,263.2	34,584.4
Finance	6,362,620	12	2,763.1	42,775.5	5,542.4	1,491.3	—	19,688.9	5,526.2	42,775.5	5,542.4	1,491.3	—	19,688.9
Average				21,216.0	84,581.7	121,916.6	68,612.0	265,824.4		21,216.0	84,581.7	121,916.6	68,612.0	265,824.4

ABA is the contribution in this paper

K= 5														
Dataset	N	D	ofv ABA	Deviation from ofv ABA [%]					cpu ABA [s]	Deviation from cpu ABA [%]				
				P-N5	P-R5	P-R50	P-R500	Rand		P-N5	P-R5	P-R50	P-R500	
Travel	5,454	24	130,871.89	0.0001	0.0000	0.0001	0.0001	-0.0192	0.002	45,104.8	34,479.4	44,869.6	129,328.6	
Npi	10,440	40	87,153.59	0.0000	0.0000	0.0000	0.0000	-0.0320	0.004	255,033.1	33,576.3	47,813.8	156,944.4	
Creditcard	30,000	24	719,975.99	0.0000	0.0000	0.0000	0.0000	-0.0160	0.010	76,110.7	36,331.0	47,208.7	119,590.6	
Adult	32,561	110	3,581,588.43	-0.0048	0.0000	0.0000	0.0000	-0.0121	0.025	48,457.0	18,036.4	32,013.6	143,843.5	
Plants	34,781	70	258,111.51	0.0000	0.0000	0.0000	0.0000	-0.0083	0.019	94,101.5	25,195.4	37,595.0	125,792.1	
Bank	45,211	53	2,396,129.86	-0.0017	0.0000	0.0000	0.0000	-0.0051	0.024	65,465.5	26,216.3	39,325.8	126,328.3	
Cifar10	50,000	3,072	9,520,045.51	0.0000	-0.0001	0.0000	0.0000	-0.0105	0.682	610,874.4	10,035.1	79,509.9	761,007.1	
Mnist	60,000	784	3,163,501.01	0.0000	0.0000	0.0000	0.0000	-0.0062	0.223	1,362,505.0	9,737.3	57,277.3	528,621.2	
Survival	110,204	4	440,812.00	-0.0038	0.0000	0.0000	0.0000	-0.0026	0.024	60,549.7	58,127.1	72,821.1	123,547.4	
Diabetes	253,680	22	5,580,938.00	-0.0003	0.0000	0.0000	0.0000	-0.0015	0.145	87,496.1	24,957.1	32,754.8	76,951.9	
Music	515,345	91	46,896,303.99	—	0.0000	0.0000	0.0000	-0.0006	0.483	—	16,723.4	30,304.8	155,567.6	
Covtype	581,012	55	31,955,602.87	-0.0005	0.0000	0.0000	0.0000	-0.0002	0.394	28,849.3	22,816.1	36,154.8	142,034.4	
Imagenet8	1,281,167	192	13,327,181.18	—	0.0000	0.0000	0.0000	-0.0003	1.872	—	12,095.5	30,794.8	203,514.9	
Imagenet32	1,281,167	3,072	265,674,709.96	—	—	—	—	-0.0003	17.733	—	—	—	—	
Census	2,458,285	68	167,163,312.00	—	0.0000	0.0000	0.0000	-0.0002	2.342	—	16,299.1	28,130.4	128,512.1	
Finance	6,362,620	12	76,351,427.75	0.0000	0.0000	0.0000	—	0.0000	4.532	24,387.9	20,434.0	26,493.8	—	
Average				-0.0009	0.0000	0.0000	0.0000	-0.0072		229,911.3	24,337.3	42,871.2	208,684.6	

ABA is the contribution in this paper

K= 5															
Dataset	N	D	Deviation from sd ABA [%]						range ABA [s]	Deviation from range ABA [%]					
			sd ABA	P-N5	P-R5	P-R50	P-R500	Rand		P-N5	P-R5	P-R50	P-R500	Rand	
Travel	5,454	24	4.8	7,667.8	3,608.7	3,631.3	4,152.6	3,082.9	13.3	7,821.7	3,518.5	3,689.9	4,207.4	3,332.3	
Npi	10,440	40	0.3	245.0	440.8	280.8	466.0	27,579.4	0.9	226.3	406.2	282.4	354.1	26,261.8	
Creditcard	30,000	24	2,475.4	114.7	115.2	103.9	122.7	208.2	6,985.7	102.2	114.5	88.0	125.3	202.9	
Adult	32,561	110	11,616.3	31.2	2.6	24.5	6.4	69.6	29,830.0	35.7	14.3	33.0	13.6	84.5	
Plants	34,781	70	0.9	997.3	635.1	823.1	947.5	66,157.8	2.9	932.1	597.4	727.5	919.6	62,448.5	
Bank	45,211	53	5,303.9	11.4	7.9	-2.4	1.7	-2.7	13,524.9	17.8	14.8	1.4	5.6	4.7	
Cifar10	50,000	3,072	38.4	19,311.3	21,843.0	27,201.3	20,001.0	23,785.2	98.9	21,663.2	23,558.9	29,167.5	20,656.8	27,983.1	
Mnist	60,000	784	3.9	5,659.0	3,781.6	3,837.8	4,381.2	31,747.0	10.6	5,838.0	3,671.2	3,822.8	4,250.1	31,659.9	
Survival	110,204	4	3.6	19,239.1	5,936.4	8,222.4	8,351.9	12,514.1	9.5	18,843.9	6,215.4	8,744.6	9,263.3	13,148.7	
Diabetes	253,680	22	6.5	40,109.3	13,297.6	12,622.8	10,853.3	43,579.7	15.6	45,253.3	15,983.7	15,023.4	12,706.3	44,789.9	
Music	515,345	91	5,872.7	—	322.2	408.4	452.0	794.0	16,478.4	—	314.6	389.2	431.7	785.1	
Covtype	581,012	55	93,564.1	13.1	1.0	1.6	1.0	37.8	192,502.9	40.4	3.0	2.8	2.0	68.1	
Imagenet8	1,281,167	192	0.8	—	227,728.0	243,564.5	451,199.2	433,347.3	2.3	—	247,409.1	242,311.0	482,542.0	454,750.8	
Imagenet32	1,281,167	3,072	8.0	—	—	—	—	568,933.2	21.9	—	—	—	—	608,410.1	
Census	2,458,285	68	68.2	—	4,940.4	5,782.9	6,532.9	50,867.9	185.3	—	5,122.2	5,648.2	6,828.8	56,229.6	
Finance	6,362,620	12	151,605.7	289.7	34.0	18.8	—	341.9	382,057.0	338.4	47.4	28.8	—	394.8	
Average				7,807.4	18,846.3	20,434.8	36,247.8	78,940.2		8,426.1	20,466.1	20,664.0	38,736.2	83,159.7	

ABA is the contribution in this paper

$K=50$

Dataset	N	D	Deviation from ofv ABA [%]						cpu ABA [s]	Deviation from cpu ABA [%]			
			ofv ABA	P-N5	P-R5	P-R50	P-R500	Rand		P-N5	P-R5	P-R50	P-R500
Travel	5,454	24	130,865.64	-0.1545	-0.0231	-0.0049	0.0018	-0.4567	0.013	6,996.7	5,207.1	7,022.0	21,125.4
Npi	10,440	40	87,151.79	-0.0030	-0.0104	-0.0032	-0.0008	-0.4710	0.017	56,553.5	7,444.7	10,988.0	39,133.6
Creditcard	30,000	24	719,975.64	-0.0019	-0.0006	-0.0002	0.0000	-0.1518	0.065	12,142.7	5,685.3	7,706.2	20,790.0
Adult	32,561	110	3,580,782.93	-0.0562	-0.0008	0.0000	0.0001	-0.1270	0.131	9,342.8	3,441.3	6,589.3	33,151.0
Plants	34,781	70	258,110.36	-0.0041	-0.0018	-0.0005	-0.0001	-0.1290	0.066	27,027.2	7,463.3	11,250.5	40,836.4
Bank	45,211	53	2,396,116.90	-0.0321	-0.0003	-0.0001	0.0000	-0.0747	0.121	12,696.2	5,248.6	8,054.1	28,001.6
Cifar10	50,000	3,072	9,519,831.08	-0.0021	-0.0047	-0.0023	-0.0011	-0.0988	3.329	123,463.4	2,329.2	19,467.6	191,611.8
Mnist	60,000	784	3,163,465.23	-0.0006	-0.0032	-0.0016	-0.0007	-0.0808	1.071	287,811.2	2,161.1	14,192.6	136,768.3
Survival	110,204	4	440,811.95	-0.0443	0.0000	0.0000	0.0000	-0.0440	0.065	22,741.0	22,756.3	28,388.3	47,036.9
Diabetes	253,680	22	5,580,937.73	-0.0042	0.0000	0.0000	0.0000	-0.0188	0.564	22,034.0	6,323.8	8,485.7	21,847.6
Music	515,345	91	46,896,303.73	—	0.0000	0.0000	0.0000	-0.0086	1.741	—	4,722.5	9,461.9	51,148.3
Covtype	581,012	55	31,955,542.41	-0.0066	0.0000	0.0000	0.0000	-0.0051	1.944	5,706.4	4,586.8	7,490.2	33,396.2
Imagenet8	1,281,167	192	13,327,181.16	—	0.0000	0.0000	0.0000	-0.0041	7.015	—	3,285.6	9,264.4	65,690.6
Imagenet32	1,281,167	3,072	265,674,690.50	—	—	—	—	-0.0038	87.515	—	—	—	—
Census	2,458,285	68	167,163,311.76	—	0.0000	0.0000	0.0000	-0.0020	7.596	—	5,081.5	9,313.4	46,120.8
Finance	6,362,620	12	76,351,393.93	-0.0004	0.0000	0.0000	—	-0.0006	24.730	4,367.9	3,695.3	4,878.0	—
Average				-0.0258	-0.0030	-0.0009	-0.0001	-0.1048		49,240.3	5,962.2	10,836.8	55,475.6

ABA is the contribution in this paper

$K=50$

Dataset	N	D	Deviation from sd ABA [%]						range ABA [s]	Deviation from range ABA [%]				
			sd ABA	P-N5	P-R5	P-R50	P-R500	Rand		P-N5	P-R5	P-R50	P-R500	Rand
Travel	5,454	24	6.5	1,508.1	1,090.1	1,075.5	1,051.6	1,099.6	25.8	1,901.8	1,228.2	1,175.2	1,399.5	1,134.5
Npi	10,440	40	2.1	85.2	106.5	71.6	79.4	1,861.7	6.6	126.8	173.9	103.2	110.1	2,563.1
Creditcard	30,000	24	1,601.8	21.0	8.2	12.5	11.6	32.0	9,277.8	17.0	6.5	16.1	8.3	27.7
Adult	32,561	110	4,117.9	56.6	32.6	31.1	38.3	72.5	29,849.6	25.9	23.6	19.8	26.9	48.5
Plants	34,781	70	1.6	772.7	334.3	307.9	388.4	14,663.5	6.7	945.4	336.0	322.9	301.3	16,335.1
Bank	45,211	53	1,941.2	11.5	-0.6	-0.9	-0.9	9.0	13,960.6	13.3	2.2	1.6	1.4	11.2
Cifar10	50,000	3,072	39.8	7,393.3	7,227.2	7,334.1	7,152.4	7,780.1	185.5	7,080.5	7,503.7	6,631.3	7,753.4	8,199.7
Mnist	60,000	784	7.8	989.9	997.1	901.8	849.2	6,670.8	35.1	1,021.3	1,029.3	815.5	836.6	6,571.0
Survival	110,204	4	4.1	4,914.0	2,108.0	2,098.2	2,324.5	4,971.3	17.2	5,126.9	2,175.7	2,326.0	2,335.8	5,232.7
Diabetes	253,680	22	13.9	5,805.1	1,814.2	2,289.1	2,291.3	7,969.0	55.7	6,280.3	1,937.8	2,860.7	2,438.7	9,484.6
Music	515,345	91	3,886.1	—	232.5	229.5	237.5	360.2	23,404.1	—	138.4	149.9	159.1	222.2
Covtype	581,012	55	45,532.1	8.6	0.4	2.5	1.9	8.9	193,064.5	24.9	4.4	5.8	5.5	24.5
Imagenet8	1,281,167	192	1.1	—	89,253.9	95,144.3	98,753.6	101,879.7	5.5	—	89,446.3	83,880.3	92,875.2	83,595.2
Imagenet32	1,281,167	3,072	14.2	—	—	—	—	133,677.2	65.5	—	—	—	—	123,419.8
Census	2,458,285	68	118.5	—	1,282.9	1,149.6	1,348.3	9,585.9	424.3	—	1,709.6	1,438.1	1,597.5	12,278.6
Finance	6,362,620	12	179,045.5	15.8	5.2	5.0	—	22.7	390,066.3	119.7	29.3	20.6	—	122.5
Average				1,798.5	6,966.2	7,376.8	8,180.5	18,166.5		1,890.3	7,049.7	6,651.1	7,846.4	16,829.4

ABA is the contribution in this paper

K= 500														
Dataset	N	D	ofv ABA	Deviation from ofv ABA [%]					cpu ABA [s]	Deviation from cpu ABA [%]				
				P-N5	P-R5	P-R50	P-R500	Rand		P-N5	P-R5	P-R50	P-R500	
Travel	5,454	24	130,132.79	-5.4124	-2.2050	-1.0382	-0.4143	-8.1005	0.223	299.9	200.2	304.4	1,121.8	
Npi	10,440	40	86,977.79	-1.0587	-1.1193	-0.5510	-0.2977	-4.5654	0.200	4,636.6	530.1	833.0	3,253.1	
Creditcard	30,000	24	719,650.34	-0.3567	-0.1237	-0.0569	-0.0294	-1.6077	1.621	376.3	129.0	213.0	747.4	
Adult	32,561	110	3,559,076.53	-0.4834	-0.0976	-0.0238	-0.0039	-0.9120	2.433	425.8	90.9	268.6	1,716.2	
Plants	34,781	70	258,012.75	-0.6240	-0.1405	-0.0633	-0.0296	-1.3518	1.043	1,623.5	366.5	628.9	2,539.2	
Bank	45,211	53	2,394,667.79	-0.5587	-0.0674	-0.0165	-0.0041	-0.7586	2.705	457.3	137.8	253.3	1,186.7	
Cifar10	50,000	3,072	9,512,381.49	-0.0720	-0.1134	-0.0612	-0.0329	-0.9384	9.352	44,271.4	773.5	7,077.4	69,480.1	
Mnist	60,000	784	3,161,729.25	-0.0893	-0.1047	-0.0543	-0.0275	-0.7819	2.952	104,310.8	721.4	5,298.9	50,525.8	
Survival	110,204	4	440,805.45	-0.4440	0.0001	0.0001	0.0001	-0.4297	0.347	4,144.9	3,981.9	5,088.4	8,748.9	
Diabetes	253,680	22	5,580,909.32	-0.0579	-0.0002	0.0000	0.0000	-0.1943	1.860	6,915.3	1,832.5	2,517.1	6,664.1	
Music	515,345	91	46,896,279.66	—	-0.0008	-0.0003	-0.0001	-0.0959	4.759	—	1,629.6	3,430.5	18,991.9	
Covtype	581,012	55	31,953,334.21	-0.0632	-0.0001	0.0000	0.0000	-0.0683	6.562	1,618.7	1,280.2	2,232.2	10,071.4	
Imagenet8	1,281,167	192	13,327,177.87	—	-0.0003	-0.0002	-0.0001	-0.0398	19.872	—	1,097.1	3,258.5	23,641.2	
Imagenet32	1,281,167	3,072	265,673,470.75	—	—	—	—	-0.0380	423.764	—	—	—	—	
Census	2,458,285	68	167,163,282.01	—	0.0000	0.0000	0.0000	-0.0205	25.156	—	1,472.1	2,770.7	14,081.4	
Finance	6,362,620	12	76,350,935.72	-0.0054	0.0000	0.0000	—	-0.0069	104.993	953.7	787.3	1,064.1	—	
Average				-0.7688	-0.2649	-0.1244	-0.0600	-1.2444		14,169.5	1,002.0	2,349.3	15,197.8	

ABA is the contribution in this paper

$K= 500$															
Dataset	N	D	Deviation from sd ABA [%]					range ABA [s]	Deviation from range ABA [%]						
			sd ABA	P-N5	P-R5	P-R50	P-R500		P-N5	P-R5	P-R50	P-R500	Rand		
Travel	5,454	24	8.7	317.2	235.1	237.7	252.0	292.8	50.0	311.1	236.3	227.7	264.2	295.2	
Npi	10,440	40	2.2	104.5	88.6	60.9	45.6	474.6	10.3	194.9	127.1	105.2	71.9	634.3	
Creditcard	30,000	24	580.5	8.2	7.1	6.2	7.3	12.3	9,485.7	2.9	2.4	3.3	3.5	6.4	
Adult	32,561	110	1,538.1	48.2	31.1	34.8	35.4	50.3	31,439.6	7.5	6.5	11.8	8.8	7.3	
Plants	34,781	70	3.7	1,203.4	131.7	77.1	79.2	2,026.7	19.5	1,647.7	155.0	76.8	75.5	2,225.3	
Bank	45,211	53	627.7	13.2	3.3	4.3	5.0	11.5	13,983.5	5.8	2.4	3.0	3.9	5.8	
Cifar10	50,000	3,072	61.6	1,495.3	1,440.2	1,455.8	1,467.9	1,474.5	386.3	1,487.3	1,385.2	1,584.0	1,456.9	1,528.9	
Mnist	60,000	784	7.8	478.9	429.8	357.4	312.0	2,020.0	56.9	428.4	349.5	284.9	223.6	1,723.8	
Survival	110,204	4	4.2	1,539.6	632.1	614.5	637.0	1,548.7	25.6	1,531.3	668.6	583.2	550.2	1,485.4	
Diabetes	253,680	22	13.7	1,883.5	624.5	606.4	602.0	2,333.5	82.4	1,839.1	638.6	628.9	659.9	2,506.5	
Music	515,345	91	2,479.9	—	80.4	70.9	86.2	137.2	27,386.5	—	35.3	30.1	28.5	69.7	
Covtype	581,012	55	14,689.7	7.8	4.9	4.7	5.5	7.7	191,779.7	9.8	5.5	5.3	7.0	8.9	
Imagenet8	1,281,167	192	1.8	—	18,607.8	18,366.9	19,336.8	19,379.7	11.1	—	18,621.6	18,174.4	21,089.8	18,772.0	
Imagenet32	1,281,167	3,072	23.5	—	—	—	—	25,377.8	139.3	—	—	—	—	25,850.3	
Census	2,458,285	68	156.9	—	339.0	327.5	319.0	2,097.0	568.9	—	635.6	572.0	582.2	3,472.5	
Finance	6,362,620	12	69,135.2	7.1	1.8	1.8	—	4.2	396,631.3	74.3	8.4	8.8	—	41.0	
Average				592.2	1,510.5	1,481.8	1,656.5	3,578.0		628.3	1,525.2	1,486.6	1,787.6	3,664.6	

ABA is the contribution in this paper

K= 1000														
Dataset	N	D	ofv ABA	Deviation from ofv ABA [%]					cpu ABA [s]	Deviation from cpu ABA [%]				
				P-N5	P-R5	P-R50	P-R500	Rand		P-N5	P-R5	P-R50	P-R500	
Travel	5,454	24	126,560.01	-11.0119	-4.9807	-2.4646	-0.9294	-15.3408	0.201	345.7	244.1	353.3	1,302.7	
Npi	10,440	40	86,399.25	-3.0555	-2.9480	-1.6556	-0.9476	-8.7383	0.198	4,705.3	544.1	843.4	3,285.6	
Creditcard	30,000	24	718,236.40	-1.0242	-0.3735	-0.1785	-0.0928	-3.0590	1.432	441.1	163.8	254.7	862.0	
Adult	32,561	110	3,528,501.69	-0.9517	-0.2866	-0.0912	-0.0178	-1.6242	2.167	462.1	112.7	298.7	1,800.9	
Plants	34,781	70	257,738.24	-1.5517	-0.3786	-0.1851	-0.0914	-2.6860	0.965	1,735.2	413.1	694.9	2,761.4	
Bank	45,211	53	2,390,566.35	-1.1955	-0.2490	-0.0754	-0.0230	-1.4508	2.441	535.0	161.2	305.4	1,324.7	
Cifar10	50,000	3,072	9,498,427.38	-0.1982	-0.2675	-0.1422	-0.0737	-1.7665	9.554	43,767.3	776.5	6,862.8	68,844.7	
Mnist	60,000	784	3,157,995.65	-0.3111	-0.2689	-0.1400	-0.0687	-1.4934	2.831	107,984.6	771.7	5,586.4	53,885.2	
Survival	110,204	4	440,792.96	-0.9112	0.0005	0.0006	0.0006	-0.8635	0.392	3,568.2	3,472.8	4,613.7	7,682.4	
Diabetes	253,680	22	5,580,808.28	-0.1361	-0.0012	-0.0001	0.0002	-0.3815	1.864	6,697.1	1,846.1	2,530.4	6,665.0	
Music	515,345	91	46,896,196.66	—	-0.0043	-0.0015	-0.0007	-0.1929	4.794	—	1,586.8	3,319.5	19,038.5	
Covtype	581,012	55	31,949,066.78	-0.1218	-0.0006	0.0000	0.0000	-0.1296	6.526	1,650.6	1,293.9	2,212.4	10,522.5	
Imagenet8	1,281,167	192	13,327,165.86	—	-0.0010	-0.0005	-0.0003	-0.0794	19.773	—	1,100.0	3,293.8	23,787.2	
Imagenet32	1,281,167	3,072	265,670,329.59	—	—	—	—	-0.0762	481.641	—	—	—	—	
Census	2,458,285	68	167,163,196.65	—	-0.0001	0.0000	0.0000	-0.0400	25.704	—	1,416.8	2,701.0	13,882.7	
Finance	6,362,620	12	76,350,416.57	-0.0117	0.0000	0.0000	—	-0.0141	106.076	948.4	781.4	1,062.8	—	
Average				-1.7067	-0.6506	-0.3289	-0.1603	-2.3710		14,403.4	979.0	2,328.9	15,403.2	

ABA is the contribution in this paper

K= 1000															
Dataset	N	D	sd ABA	Deviation from sd ABA [%]					range ABA [s]	Deviation from range ABA [%]					
				P-N5	P-R5	P-R50	P-R500	Rand		P-N5	P-R5	P-R50	P-R500	Rand	
Travel	5,454	24	11.8	133.2	116.7	127.3	143.1	125.6	64.6	160.3	132.8	130.2	151.5	143.5	
Npi	10,440	40	3.4	71.0	55.1	38.6	35.1	182.2	13.7	142.7	89.3	49.4	38.1	316.0	
Creditcard	30,000	24	417.8	5.8	5.3	5.1	7.4	10.3	9,403.5	1.8	2.4	1.8	3.2	4.1	
Adult	32,561	110	1,182.6	37.3	29.4	31.9	29.9	36.1	31,321.5	6.5	7.1	11.5	4.1	3.9	
Plants	34,781	70	3.9	919.9	109.1	59.6	41.9	1,308.3	22.7	1,088.8	125.5	90.9	39.3	1,348.8	
Bank	45,211	53	442.7	13.2	4.7	5.6	4.3	10.5	13,868.6	4.0	1.6	2.7	1.4	3.2	
Cifar10	50,000	3,072	68.4	936.0	892.8	898.6	881.2	912.7	491.5	880.8	753.9	766.6	813.3	850.1	
Mnist	60,000	784	8.6	416.7	320.2	247.8	209.1	1,227.8	62.2	417.3	307.0	212.0	168.5	1,142.4	
Survival	110,204	4	4.7	930.1	357.6	351.1	358.1	901.2	27.2	1,071.0	417.6	404.6	401.3	1,033.2	
Diabetes	253,680	22	13.6	1,344.8	436.3	433.1	441.0	1,644.0	100.2	1,117.0	396.8	371.8	381.1	1,392.7	
Music	515,345	91	2,035.6	—	59.1	52.2	60.1	112.4	27,868.2	—	22.1	15.0	19.8	52.0	
Covtype	581,012	55	10,562.4	5.6	3.0	4.1	4.5	5.8	192,939.2	5.4	2.4	5.1	5.1	5.7	
Imagenet8	1,281,167	192	2.1	—	11,139.5	11,031.9	11,627.7	12,072.1	12.6	—	12,044.5	12,194.3	12,508.0	13,646.9	
Imagenet32	1,281,167	3,072	30.3	—	—	—	—	13,935.2	172.9	—	—	—	—	15,601.8	
Census	2,458,285	68	148.3	—	261.6	255.0	256.3	1,541.4	631.5	—	427.1	411.7	416.2	2,670.4	
Finance	6,362,620	12	49,535.3	2.2	1.4	1.7	—	1.6	397,208.7	7.8	5.7	6.9	—	7.5	
Average				401.3	919.5	902.9	1,007.1	2,126.7		408.6	982.4	978.3	1,067.9	2,388.9	

ABA is the contribution in this paper

$K=2000$

Dataset	N	D	ofv ABA	Deviation from ofv ABA [%]					cpu ABA [s]	Deviation from cpu ABA [%]			
				P-N5	P-R5	P-R50	P-R500	Rand		P-N5	P-R5	P-R50	P-R500
Travel	5,454	24	111,341.47	-18.9230	-8.4894	-3.1904	0.0038	-25.3482	0.154	483.2	343.5	488.4	1,637.2
Npi	10,440	40	84,068.50	-7.3996	-6.6095	-4.0046	-2.3669	-16.0867	0.162	5,755.6	689.4	1,055.6	4,070.1
Creditcard	30,000	24	713,230.74	-2.5618	-0.9355	-0.4338	-0.2082	-5.7919	1.200	567.4	207.8	323.9	1,054.5
Adult	32,561	110	3,453,734.26	-1.6754	-0.5862	-0.1953	-0.0291	-2.6783	2.017	509.8	126.6	310.9	1,908.7
Plants	34,781	70	256,580.84	-3.4916	-0.8754	-0.3901	-0.1582	-5.2165	0.952	1,785.6	426.3	687.6	2,806.9
Bank	45,211	53	2,375,555.55	-2.3490	-0.7139	-0.2658	-0.0889	-2.6920	2.058	654.1	219.1	365.4	1,592.3
Cifar10	50,000	3,072	9,461,043.79	-0.5772	-0.6167	-0.3223	-0.1596	-3.3886	9.292	44,570.3	797.5	7,125.9	70,149.1
Mnist	60,000	784	3,146,991.10	-0.9003	-0.6618	-0.3462	-0.1683	-2.8070	2.872	106,487.7	777.4	5,553.4	52,544.0
Survival	110,204	4	440,768.00	-1.8185	-0.0022	0.0015	0.0015	-1.7319	0.470	2,991.8	3,032.2	3,807.0	6,438.7
Diabetes	253,680	22	5,580,547.04	-0.3258	-0.0100	-0.0015	0.0000	-0.7683	1.889	6,663.6	1,822.2	2,473.6	6,576.8
Music	515,345	91	46,895,811.88	—	-0.0209	-0.0075	-0.0036	-0.3858	4.880	—	1,614.8	3,253.2	18,657.9
Covtype	581,012	55	31,936,029.61	-0.2285	-0.0022	-0.0001	0.0000	-0.2420	6.505	1,644.7	1,287.7	2,216.7	10,192.5
Imagenet8	1,281,167	192	13,327,115.17	—	-0.0035	-0.0016	-0.0009	-0.1543	20.598	—	1,057.6	3,168.9	23,086.9
Imagenet32	1,281,167	3,072	265,660,317.25	—	—	—	—	-0.1519	457.806	—	—	—	—
Census	2,458,285	68	167,162,962.58	—	-0.0004	-0.0002	-0.0001	-0.0808	26.846	—	1,378.1	2,585.8	13,237.1
Finance	6,362,620	12	76,349,371.76	-0.0247	0.0000	0.0000	—	-0.0280	107.516	926.6	773.3	1,050.3	—
Average				-3.3563	-1.3018	-0.6105	-0.2270	-4.2220		14,420.1	970.2	2,297.8	15,282.3

ABA is the contribution in this paper

$K=2000$

Dataset	N	D	sd ABA	Deviation from sd ABA [%]					range ABA [s]	Deviation from range ABA [%]				
				P-N5	P-R5	P-R50	P-R500	Rand		P-N5	P-R5	P-R50	P-R500	Rand
Travel	5,454	24	14.8	31.6	36.7	51.7	62.2	20.7	79.5	46.1	46.0	49.8	57.8	36.5
Npi	10,440	40	3.4	54.9	32.4	22.3	17.9	103.0	14.3	122.3	93.9	56.1	36.7	190.6
Creditcard	30,000	24	294.5	4.9	3.9	4.1	5.3	5.8	9,093.7	1.5	0.8	1.6	1.8	1.1
Adult	32,561	110	881.2	26.7	26.0	34.6	39.7	25.0	30,490.2	4.7	2.8	4.2	6.5	2.3
Plants	34,781	70	5.6	464.8	59.4	19.4	13.2	595.3	27.9	620.3	105.5	56.1	37.8	694.0
Bank	45,211	53	309.7	13.6	8.4	7.7	8.1	11.7	13,602.3	3.4	2.6	2.3	2.2	3.0
Cifar10	50,000	3,072	76.4	550.0	533.2	531.7	543.1	541.6	517.9	522.0	516.6	523.1	546.7	526.4
Mnist	60,000	784	9.0	370.1	253.7	185.8	149.5	777.4	68.3	334.8	199.5	144.5	114.4	607.3
Survival	110,204	4	5.3	528.5	191.7	186.3	182.8	519.5	28.9	704.3	259.4	260.4	252.2	651.3
Diabetes	253,680	22	13.3	955.2	309.1	288.8	291.1	1,172.2	115.7	700.7	242.9	206.5	214.3	958.4
Music	515,345	91	1,630.8	—	44.1	41.3	44.2	83.2	28,283.6	—	15.8	10.4	10.5	27.5
Covtype	581,012	55	7,525.9	5.4	2.8	3.5	3.0	5.5	192,621.1	4.9	2.1	3.1	2.5	5.3
Imagenet8	1,281,167	192	2.6	—	6,451.0	6,446.5	6,688.0	6,821.4	16.3	—	7,208.8	7,099.5	7,254.7	7,629.2
Imagenet32	1,281,167	3,072	36.9	—	—	—	—	8,269.2	243.3	—	—	—	—	9,009.5
Census	2,458,285	68	122.2	—	225.6	223.3	218.2	1,298.2	701.1	—	274.2	264.9	269.6	1,614.2
Finance	6,362,620	12	35,335.2	1.6	0.8	1.0	—	1.3	397,950.9	5.4	3.7	3.9	—	4.0
Average				250.6	545.2	536.5	590.5	1,265.7		255.9	598.3	579.1	629.1	1,372.5

ABA is the contribution in this paper

K= 5000														
Dataset	N	D	ofv ABA	Deviation from ofv ABA [%]					cpu ABA [s]	Deviation from cpu ABA [%]				
				P-N5	P-R5	P-R50	P-R500	Rand		P-N5	P-R5	P-R50	P-R500	
Npi	10,440	40	67,221.83	-16.7954	-13.5452	-7.2405	-3.0945	-32.5496	0.121	7,786.3	956.1	1,350.1	4,402.7	
Creditcard	30,000	24	688,499.29	-7.4919	-2.9883	-1.2745	-0.4467	-12.7936	0.892	792.0	324.9	472.0	1,463.9	
Adult	32,561	110	3,195,721.47	-3.3696	-1.2520	-0.4247	0.0147	-5.1587	1.883	581.6	146.2	390.8	2,426.2	
Plants	34,781	70	249,858.02	-8.9426	-2.3712	-0.7989	-0.1183	-11.3736	0.774	2,218.2	543.5	909.7	3,551.5	
Bank	45,211	53	2,300,020.22	-5.1379	-2.1135	-0.9595	-0.3779	-5.7099	2.074	635.8	205.9	371.6	1,598.2	
Cifar10	50,000	3,072	9,306,096.15	—	—	—	—	-7.9227	10.286	—	—	—	—	
Mnist	60,000	784	3,095,414.89	-2.8785	-1.9690	-1.0435	-0.4876	-6.3092	3.131	97,157.0	689.5	5,102.5	48,915.5	
Survival	110,204	4	439,737.17	-4.2773	-0.0091	0.0000	0.0001	-4.1807	0.648	2,188.6	2,065.7	2,650.6	4,703.9	
Diabetes	253,680	22	5,578,352.45	-0.9738	-0.1019	-0.0171	-0.0015	-1.9047	2.052	6,037.8	1,652.0	2,286.3	6,089.2	
Music	515,345	91	46,891,810.38	—	-0.1322	-0.0563	-0.0287	-0.9566	5.259	—	1,482.1	3,142.2	18,329.0	
Covtype	581,012	55	31,878,267.62	-0.5072	-0.0139	-0.0015	0.0000	-0.5263	7.047	1,505.9	1,194.5	2,036.8	9,575.6	
Imagenet8	1,281,167	192	13,326,770.00	—	-0.0169	-0.0078	-0.0040	-0.3863	23.098	—	917.1	2,721.2	19,639.9	
Imagenet32	1,281,167	3,072	265,611,922.65	—	—	—	—	-0.3670	445.812	—	—	—	—	
Census	2,458,285	68	167,160,589.77	—	-0.0027	-0.0010	-0.0004	-0.2010	29.901	—	1,223.1	2,334.2	12,010.2	
Finance	6,362,620	12	76,345,590.19	-0.0647	0.0000	0.0000	—	-0.0704	116.939	853.4	706.5	951.8	—	
Average				-5.0439	-1.8858	-0.9096	-0.3787	-6.0273		11,975.7	931.3	1,901.5	11,058.8	

ABA is the contribution in this paper

K= 5000														
Dataset	N	D	Deviation from sd ABA [%]						range ABA [s]	Deviation from range ABA [%]				
			sd ABA	P-N5	P-R5	P-R50	P-R500	Rand		P-N5	P-R5	P-R50	P-R500	Rand
Npi	10,440	40	3.8	7.0	2.5	1.7	0.3	4.6	18.7	38.4	23.8	17.9	11.6	38.7
Creditcard	30,000	24	173.5	2.7	3.0	3.7	5.2	1.9	8,153.5	0.4	1.6	0.9	1.2	-0.3
Adult	32,561	110	574.4	11.1	16.6	15.2	15.8	9.8	27,962.3	1.6	4.0	1.0	2.4	-1.0
Plants	34,781	70	8.4	148.5	7.0	-20.2	-28.4	164.9	39.1	169.6	49.8	11.9	-0.1	172.2
Bank	45,211	53	188.7	11.0	6.7	8.7	10.0	8.2	12,847.5	1.2	-0.4	2.0	2.4	0.2
Cifar10	50,000	3,072	86.8	—	—	—	—	244.6	599.8	—	—	—	—	281.7
Mnist	60,000	784	9.9	244.1	154.1	111.9	87.1	398.5	92.3	178.9	92.6	57.7	48.2	275.5
Survival	110,204	4	5.1	303.7	133.4	131.6	128.4	306.6	29.0	379.2	184.1	177.3	166.8	401.7
Diabetes	253,680	22	13.3	591.8	206.9	179.4	190.3	719.2	124.1	460.6	171.5	142.1	142.4	513.1
Music	515,345	91	1,186.0	—	30.7	31.3	33.2	57.0	28,426.1	—	11.6	9.1	15.0	21.4
Covtype	581,012	55	4,794.1	3.3	2.9	3.3	3.8	3.6	191,723.8	1.7	2.8	2.2	1.7	2.7
Imagenet8	1,281,167	192	3.2	—	3,181.7	3,195.7	3,288.8	3,382.3	23.1	—	3,298.1	3,260.6	3,785.9	3,318.7
Imagenet32	1,281,167	3,072	46.3	—	—	—	—	4,012.4	315.7	—	—	—	—	4,243.0
Census	2,458,285	68	176.2	—	108.7	106.3	108.1	521.0	793.1	—	188.3	169.5	167.4	952.5
Finance	6,362,620	12	22,507.8	0.9	0.4	0.3	—	0.9	398,104.7	2.6	2.0	2.4	—	3.1
Average				132.4	296.5	289.9	320.2	655.7		123.4	310.0	296.5	362.1	681.6

ABA is the contribution in this paper